

Name: \_\_\_\_\_ Partner: \_\_\_\_\_ Period: \_\_\_\_\_

## Context: Modeling Periodic Real-World Data

Sinusoidal functions appear in temperature records, tidal data, daylight hours, and many other natural phenomena. In this activity your group will build and evaluate a sinusoidal model from real-world data. Choose **one tier** based on your readiness. Everyone should be ready to explain the reasoning behind each parameter your group chooses.

## Tier R — Reinforce

**Context:** The table below gives the average monthly temperature ( $^{\circ}\text{F}$ ) in a small coastal town. The maximum and minimum are already labeled for you.

| Month $x$                       | 1  | 2  | 3  | 4  | 5  | 6  | 7         | 8  | 9  | 10 | 11 | 12        |
|---------------------------------|----|----|----|----|----|----|-----------|----|----|----|----|-----------|
| Temp $T$ ( $^{\circ}\text{F}$ ) | 48 | 51 | 58 | 65 | 72 | 78 | <b>80</b> | 79 | 72 | 63 | 54 | <b>46</b> |

**Step 1 — Amplitude and Midline.** The labeled max is **80** (month 7) and min is **46** (month 12 / month 1 area).

$$A = \frac{80 - 46}{2} = \underline{\hspace{2cm}} \quad D = \frac{80 + 46}{2} = \underline{\hspace{2cm}}$$

**Step 2 — Period and  $B$ .** The maximum occurs at month 7. The next maximum would occur at month \_\_\_\_\_ (one year later), so the period  $k = \underline{\hspace{2cm}}$ .

$$B = \frac{2\pi}{k} = \underline{\hspace{2cm}}$$

**Step 3 — Phase Shift.** Using a cosine model,  $C = x_{\max} = \underline{\hspace{2cm}}$ .

**Step 4 — Write the Model.**  $T(x) = A \cos(B(x - C)) + D = \underline{\hspace{4cm}}$

**Step 5 — Validate.** Predict  $T(1)$  using your model:  $T(1) \approx \underline{\hspace{2cm}}$ . Actual value from the table: 48. How close is your prediction?  $\underline{\hspace{4cm}}$

**Step 6 — Predict.** Use your model to estimate the average temperature in month 4.5 (mid-May).  $T(4.5) \approx \underline{\hspace{2cm}}^{\circ}\text{F}$ .

# Tier A — Apply

**Context:** The table below shows the height of the tide (in feet above mean sea level) recorded every hour at a harbor over one tidal cycle.

|             |     |     |     |     |     |     |     |      |      |      |      |      |     |
|-------------|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|-----|
| $t$ (hr)    | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7    | 8    | 9    | 10   | 11   | 12  |
| $h(t)$ (ft) | 1.5 | 3.8 | 5.5 | 6.2 | 5.5 | 3.8 | 1.5 | -0.8 | -2.5 | -3.2 | -2.5 | -0.8 | 1.5 |

- Identify the maximum and minimum values and the times at which they occur.

Max = \_\_\_\_\_ at  $t =$  \_\_\_\_\_ hr.    Min = \_\_\_\_\_ at  $t =$  \_\_\_\_\_ hr.

- Compute  $A$  and  $D$ .

$$A = \frac{\max - \min}{2} = \underline{\hspace{2cm}} \quad D = \frac{\max + \min}{2} = \underline{\hspace{2cm}}$$

What does the midline represent in the context of tidal data? \_\_\_\_\_

- Estimate the period  $k$  from the data. Explain how you found it. \_\_\_\_\_

$$k = \underline{\hspace{2cm}} \text{ hr} \quad \Rightarrow \quad B = \frac{2\pi}{k} = \underline{\hspace{2cm}}$$

- Using a cosine model with  $C = t_{\max}$ , write the full sinusoidal model.

$$h(t) = \underline{\hspace{4cm}}$$

- Validate: compute  $h(0)$  from your model and compare to the actual value.

$$h(0) = \underline{\hspace{2cm}} \quad \text{Actual: 1.5 ft} \quad \text{Residual: } \underline{\hspace{2cm}}$$

- A ship requires a minimum tide height of 2 ft to enter the harbor safely. Using your model, at approximately what times during the 12-hour period is entry safe? \_\_\_\_\_

\_\_\_\_\_

\_\_\_\_\_

## Tier E — Extend

**Context:** The tables below give average monthly high temperatures for two cities, City A (coastal) and City B (inland continental).

**City A:**

| Month     | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|-----------|----|----|----|----|----|----|----|----|----|----|----|----|
| Temp (°F) | 58 | 60 | 63 | 66 | 69 | 72 | 74 | 75 | 73 | 68 | 63 | 59 |

**City B:**

| Month     | 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 |
|-----------|----|----|----|----|----|----|----|----|----|----|----|----|
| Temp (°F) | 28 | 34 | 50 | 66 | 78 | 88 | 94 | 90 | 76 | 60 | 42 | 31 |

1. For each city, compute  $A$ ,  $D$ ,  $k$ ,  $B$ , and  $C$  (using cosine), then write the sinusoidal model. \_\_\_\_\_

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2. Compare your models. Which city has the larger amplitude? What does this mean physically? \_\_\_\_\_

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3. For each city, compute the residual at  $x = 1$  (January). Which model fits January better? \_\_\_\_\_

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4. Compute residuals at  $x = 4, 7, 10$  for City B. Based on the residuals, do you think a sinusoidal model is a good fit for City B? Explain. \_\_\_\_\_

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5. State the contextual domain for each model. Using City B's model, predict the average high temperature in month 14. Is this prediction trustworthy? Explain. \_\_\_\_\_

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